INSTITUTE FOR SOCIAL RESEARCH
UNIVERSITY OF MICHIGAN

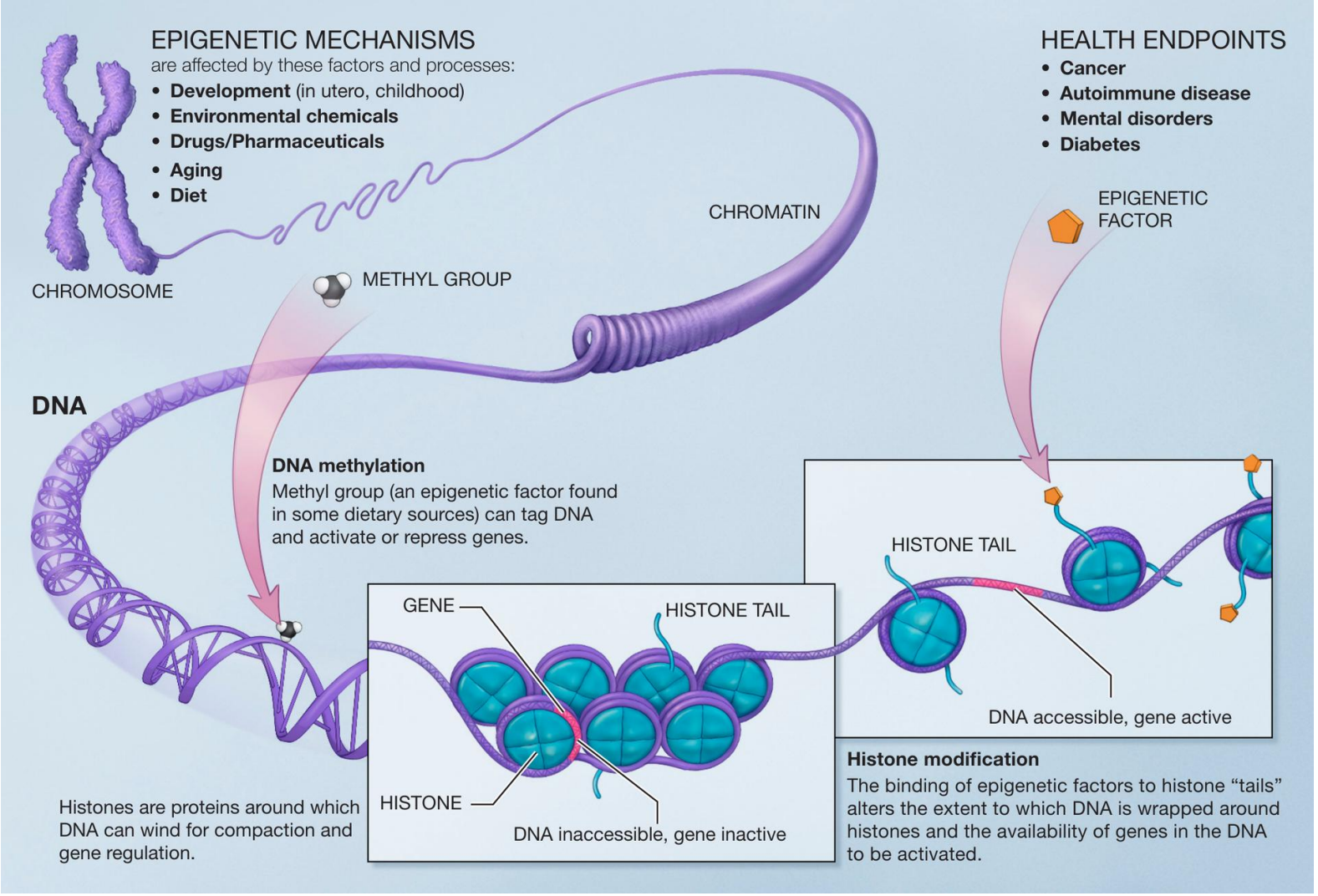
Body Mass Index (BMI) DNA Methylation Scores in the Health and Retirement Study

Trey Smith

treysm@umich.edu

GSA 2023

DNA Methylation (DNAm)



- The majority of DNA methylation occurs on cytosines that precede a guanine nucleotide or "CpG sites"
- Epigenome-Wide Association Studies (EWAS) as well as various machine learning techniques are used to select CpGs associated with various outcomes

Research Question

- DNA Methylation scores have been shown to predict outcomes over and above the phenotype they are created to predict

- Do DNAm BMI scores from CpGs identified in previous studies of BMI and methylation predict health outcomes over and above current BMI in HRS?

Health and Retirement Study (HRS)

- The University of Michigan Health and Retirement Study (HRS) is a longitudinal panel study that surveys a representative sample of approximately 20,000 people in America, supported by the National Institute on Aging (NIA U01AG009740) and the Social Security Administration.
- DNA methylation assays were done on a non-random subsample (n=4,104) people who participated in the 2016 Venous Blood Study. This subsample, once weighted, fully represents the entire HRS sample. A total of 4,018 samples passed quality control (QC).

3 different BMI scores

- Wahl (187 CpGs) - EWAS Meta-Analysis of 5,387 individuals from the EPICOR (N=514), KORA (N=2,193) and LOLIPOP (N=2,680) population studies
- McCartney (1109 CpGs) – LASSO Penalized Regression Model of 5,087 individuals from Generation Scotland (GS)
- Hamilton (435 CpGs) – LASSO Penalized Regression Model of 2,562 individuals from Generation Scotland (GS)
- Both McCartney and Hamilton were tested in the Lothian Birth Cohort (LBC)

Chronic Conditions in HRS

1) HIGH BLOOD PRESSURE
2) DIABETES
3) CANCER OF ANY KIND EXCLUDING SKIN
4) LUNG DISEASE
5) HEART CONDITION
6) STROKE
7) EMOTIONAL/PSYCHIATRIC PROBLEMS
8) ARTHRITIS
9) TOLD BY HEALTH PROFESSIONAL HAVE SLEEP DISORDER

Models (ran separately by sex)

- **Model 1:** variable of interest ~ epigenetic BMI score
- **Model 2:** variable of interest ~ epigenetic BMI score + Demographics (Age, Race, Ethnicity)
- **Model 3**: variable of interest ~ epigenetic BMI score + Demographics + SES (Education, Wealth)
- **Model 4:** variable of interest ~ epigenetic BMI score + Demographics + SES + Health Behaviors (Drinking, Smoking, **Obesity**)
- **Model 5:** variable of interest ~ epigenetic BMI score + Demographics + SES + Health Behaviors + Cell Type % (CD8N, NK, MONO, B, CD8T)

Cross-sectional in 2016 HRS

Outcome	BMI Wahl	BMI McCartney	BMI Hamilton
Chronic Conditions	M ***, W ***	M ***, W ***	M ***, W ***
ADL/IADLs	M *		
HDL Cholesterol	M ***, W ***	M ***, W *	M ***, W ***
LDL Cholesterol	W *		M *, W **
Triglycerides	M **, W **	M ***, W *	M **, W ***

***p<0.001; **p<0.01; *p<0.05

M = Men, W = Women

Incident conditions in 2018 HRS

Outcome	BMI Wahl	BMI McCartney	BMI Hamilton
Mortality			
Diabetes	W ***	M ***, W ***	W ***
Psychiatric problems	M **	M ***, W **	M ***
High Blood Pressure			M *
Sleep Disorder	W*	W **	W ***

***p<0.001; **p<0.01; *p<0.05

Incident conditions in 2020 HRS

Outcome	BMI Wahl	BMI McCartney	BMI Hamilton
Mortality		W*	W*
Diabetes	W**	W*	
Psychiatric Problems		M*	M*
Arthritis	M*		
Lung Disease	M*, W**		
Stroke		M***	
Heart Condition	M***		

***p<0.001; **p<0.01; *p<0.05

Future Analyses

- Sensitivity analyses adjusting for other epigenetic measures (e.g. GrimAge)
- Models with BMI as the predictor adjusting for DNAm scores
- Test more specific outcomes (heart attack, sleep apnea, etc)
- Look at ever having the condition, not just incident in 2018 and 2020
- Relationship between BMI over the life course and DNAm scores

Discussion

- These scores are associated with various aspects of metabolic syndrome (high blood pressure, high blood sugar, excess body fat around the waist, and abnormal cholesterol levels).
- Conditions which increase in prevalence during aging, such as obesity, insulin resistance, inflammation, stress and hypertension also contribute to increase prevalence of metabolic syndrome.
- Metabolic syndrome accelerates aging, the development of insulin resistance and cardiovascular disease.
- Can we take the CpGs from these scores and train a DNA Methylation surrogate to predict metabolic syndrome?
- How would this compare to DNAm measures of age acceleration such as GrimAge?

References

1. Hamilton OKL, Zhang Q, McRae AF, Walker RM, Morris SW, Redmond P, Campbell A, Murray AD, Porteous DJ, Evans KL, McIntosh AM, Deary IJ, Marioni RE. An epigenetic score for BMI based on DNA methylation correlates with poor physical health and major disease in the Lothian Birth Cohort. Int J Obes (Lond). 2019 Sep;43(9):1795-1802. doi: 10.1038/s41366-018-0262-3. Epub 2019 Mar 6. PMID: 30842548; PMCID: PMC6760607.

2. McCartney DL, Hillary RF, Stevenson AJ, Ritchie SJ, Walker RM, Zhang Q, Morris SW, Bermingham ML, Campbell A, Murray AD, Whalley HC, Gale CR, Porteous DJ, Haley CS, McRae AF, Wray NR, Visscher PM, McIntosh AM, Evans KL, Deary IJ, Marioni RE. Epigenetic prediction of complex traits and death. Genome Biol. 2018 Sep 27;19(1):136. doi: 10.1186/s13059-018-1514-1. PMID: 30257690; PMCID: PMC6158884.

3. Wahl S, Drong A, Lehne B, Loh M, Scott WR, Kunze S, Tsai PC, Ried JS, Zhang W, Yang Y, Tan S, Fiorito G, Franke L, Guarrera S, Kasela S, Kriebel J, Richmond RC, Adamo M, Afzal U, Ala-Korpela M, Albetti B, Ammerpohl O, Apperley JF, Beekman M, Bertazzi PA, Black SL, Blancher C, Bonder MJ, Brosch M, Carstensen-Kirberg M, de Craen AJ, de Lusignan S, Dehghan A, Elkalaawy M, Fischer K, Franco OH, Gaunt TR, Hampe J, Hashemi M, Isaacs A, Jenkinson A, Jha S, Kato N, Krogh V, Laffan M, Meisinger C, Meitinger T, Mok ZY, Motta V, Ng HK, Nikolakopoulou Z, Nteliopoulos G, Panico S, Pervjakova N, Prokisch H, Rathmann W, Roden M, Rota F, Rozario MA, Sandling JK, Schafmayer C, Schramm K, Siebert R, Slagboom PE, Soininen P, Stolk L, Strauch K, Tai ES, Tarantini L, Thorand B, Tigchelaar EF, Tumino R, Uitterlinden AG, van Duijn C, van Meurs JB, Vineis P, Wickremasinghe AR, Wijmenga C, Yang TP, Yuan W, Zhernakova A, Batterham RL, Smith GD, Deloukas P, Heijmans BT, Herder C, Hofman A, Lindgren CM, Milani L, van der Harst P, Peters A, Illig T, Relton CL, Waldenberger M, Järvelin MR, Bollati V, Soong R, Spector TD, Scott J, McCarthy MI, Elliott P, Bell JT, Matullo G, Gieger C, Kooner JS, Grallert H, Chambers JC. Epigenome-wide association study of body mass index, and the adverse outcomes of adiposity. Nature. 2017 Jan 5;541(7635):81-86. doi: 10.1038/nature20784. Epub 2016 Dec 21. PMID: 28002404; PMCID: PMC5570525.

4. Moore, L., Le, T. & Fan, G. DNA Methylation and Its Basic Function. *Neuropsychopharmacol* **38**, 23–38 (2013). https://doi.org/10.1038/npp.2012.112

5. Crimmins EM, Faul J, Thyagarajan B, Weir D. 2017. Venous blood collection and assay protocol in the 2016 Health and Retirement Study 2016 Venous Blood Study (VBS). http://hrsonline.isr.umich.edu/modules/meta/ vbs/2016/desc/HRS2016VBSDD.pdf.

6. Bonomini F, Rodella LF, Rezzani R. Metabolic syndrome, aging and involvement of oxidative stress. Aging Dis. 2015 Mar 10;6(2):109-20. doi: 10.14336/AD.2014.0305. PMID: 25821639; PMCID: PMC4365955.

DISCLOSURE(S)

I have no commercial relationships to disclose.